

Dynamics and Reconstruction in the Observable Measure

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Abstract

Starting from a σ -additive probability measure on the algebra of discriminable events [11], the Rokhlin disintegration theorem forces a conditional regularity kernel and a minimal sufficient factor Q_* , the coarsest reduction carrying full conditional information. Indexing over time, temporal coherence forces the Chapman–Kolmogorov equation, yielding a Markov semigroup $\{\Pi_t\}$ and Koopman–Perron duality; the dynamics of observation and the underlying state space are derived, not assumed. In a measure-preserving system Q_* becomes the delay map Φ_h , and reconstruction — that $\mathcal{O}_h = \mathcal{B} \bmod \mu$, density of the delay algebra in L^2 , and injectivity of Φ_h are equivalent — is the faithfulness question about this factor. The Stone space [11] built to derive σ -additivity is identified with X itself, up to measure-theoretic isomorphism, when reconstruction holds.

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1 Introduction

Koopman studied dynamical systems through observables [8], but the dynamical law T remains primitive: the operator $U_T f = f \circ T$ is defined only after an underlying evolution has already been specified. The classical stochastic-process tradition proceeds similarly. Prediction is formulated relative to a filtration, so temporal order and information flow are built into the formalism from the outset [3]. That ordering is natural, but it need not be conceptually prior. One may instead ask whether observable probability already carries a conditional regularity between observables, and whether the familiar semigroup and predictive structures arise only when that regularity is organised coherently over time.

In [11], σ -additivity is not assumed but derived: a probability measure is forced into existence by the coherence of finite observations, at the cost of a single admissibility condition that no first-order structural requirement can supply. The measure is earned, not given. It does, however, already carry more than it was built to carry. Any two observations Q and F — no hypothesis on time or order — determine a conditional regularity of F given Q : the Markov kernel $\kappa_Q(q, \cdot) = P(F \in \cdot \mid Q = q)$, which the Rokhlin disintegration theorem [12] forces into existence — it need not be separately posited. Composing with Q gives the minimal sufficient factor Q_* , the coarsest reduction of the sample point that carries full conditional information about F . Temporal prediction is the special case in which F is a future observable; nothing in the construction requires it.

Index the conditional regularity over time and the kernels $\{\Pi_t\}$ must compose consistently. That coherence forces Chapman–Kolmogorov — derived, not assumed — and from it a Markov semigroup and Koopman–Perron duality $\int K_t g d\mu = \int g d(\mu \star \Pi_t)$. In a measure-preserving system with $Q_t = h \circ T^t$, the factor Q_* becomes the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$, and the question becomes internal: does this factor see all of X , or only a proper quotient? The

answer — that $\mathcal{O}_h = \mathcal{B} \bmod \mu$, density of the delay algebra in L^2 , and injectivity of Φ_h are all equivalent — is the reconstruction theorem. The Stone space [11] built to derive σ -additivity is identified with X itself when reconstruction holds.

2 Conditional regularity and the canonical factor

2.1 The conditional regularity kernel

Let $(\Omega, \mathcal{F}, P)$ be a σ -additive probability space on an observable algebra [11], and let Q, F be two measurable maps:

- $Q : \Omega \rightarrow \alpha$, where $(\alpha, \mathcal{A})$ is a measurable space;
- $F : \Omega \rightarrow \beta$, where $(\beta, \mathcal{B})$ is a standard Borel space.

The standard Borel hypothesis on β is used once: it is the hypothesis under which the Rokhlin disintegration theorem [12] guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write $\mathcal{P}(\beta)$ for the space of probability measures on β , equipped with its canonical (Giry) σ -algebra. The joint pushforward $\rho := P \circ (Q, F)^{-1}$ on $\alpha \times \beta$ has marginals $\rho_\alpha = P \circ Q^{-1}$ and $\rho_\beta = P \circ F^{-1}$.

The regular conditional distribution of F given Q is the unique (up to $P \circ Q^{-1}$ -null sets) Markov kernel $\kappa_Q : \alpha \rightarrow \mathcal{P}(\beta)$ satisfying

$$\int_A \kappa_Q(q, B) d(P \circ Q^{-1})(q) = P(Q \in A, F \in B)$$

for all measurable $A \subseteq \alpha$ and $B \subseteq \beta$.

Definition 2.1 (Conditional regularity kernel). The *conditional regularity kernel* of F given Q is κ_Q , the disintegration kernel of $\rho = P \circ (Q, F)^{-1}$ over its first marginal: $\kappa_Q(q, A) = P(F \in A \mid Q = q)$. When F is a future observable the kernel describes conditional prediction; the object itself is neutral on time.

The kernel is covariant under measurable maps: if $\pi : \alpha \rightarrow \alpha'$ is measurable and $Q' = \pi \circ Q$, then $\kappa_Q(q, \cdot) = \kappa_{Q'}(\pi(q), \cdot)$ for $(P \circ Q^{-1})$ -almost every q , by uniqueness of disintegration.

Composing κ_Q with Q gives the map sending each sample point to its conditional regularity.

2.2 The minimal sufficient factor

Definition 2.2 (Minimal sufficient factor). The *minimal sufficient factor* is

$$Q_* : \Omega \rightarrow \mathcal{P}(\beta), \quad Q_*(\omega) = \kappa_Q(Q(\omega), \cdot).$$

Since κ_Q is a Markov kernel it is measurable, and Q_* inherits measurability by composition.

Theorem 2.3 (Factorization via the canonical factor). *For every bounded measurable $g : \beta \rightarrow \mathbb{R}$:*

$$E[g(F) \mid \sigma(Q)] = \int g dQ_*(\cdot) \quad P\text{-a.e.}$$

Proof. Let $h(\omega) := \int g dQ_*(\omega)$. We verify the two defining properties.

Measurability. The map $q \mapsto \int g d\kappa_Q(q, \cdot)$ is strongly measurable; since $Q_* = \kappa_Q(Q(\cdot), \cdot)$, the map h is $\sigma(Q)$ -measurable.

Set-integral identity. For $S = Q^{-1}(A)$:

$$\begin{aligned} \int_S h dP &= \int_A \int g d\kappa_Q(q, \cdot) d(P \circ Q^{-1})(q) \quad (\text{pushforward}) \\ &= \int_{A \times \beta} g(f) d\rho(q, f) \quad (\text{disintegration}) \\ &= \int_S g(F) dP. \end{aligned}$$

The conclusion follows from a.e. uniqueness of conditional expectation. $\square$

Since $\sigma(Q_*) \subseteq \sigma(Q)$ and the right-hand side is already $\sigma(Q_*)$ -measurable, Q_* is a sufficient statistic for F given Q : $E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$ P -a.e. It is *minimal* in the sense that every sufficient statistic factors through it: Q_* is the coarsest reduction of ω that carries the full conditional regularity of F given Q . The operator $(K_{Q_*}g)(q_*) = \int g dq_*$ on $\mathcal{P}(\beta)$ recasts this as $E[g(F) \mid \sigma(Q)] = (K_{Q_*}g) \circ Q_*$ P -a.e.

3 Dynamics and duality

3.1 Markov semigroup

When the conditional regularity kernel κ_Q is indexed over a time parameter, the kernels cannot be chosen independently: the regularity of F at time $t + s$ given Q at time 0 must be consistent with first observing at time s and then at time t . This consistency requirement forces the Chapman–Kolmogorov equation.

Concretely, index over a discrete time horizon $\mathbb{N}$ and let $\Pi_t : \gamma \rightarrow \mathcal{P}(\gamma)$ be a Markov kernel on a state space γ , with $\Pi_0(q, \cdot) = \delta_q$. Temporal coherence of the conditional regularity demands

$$\Pi_{t+s}(q, \cdot) = \int_{\gamma} \Pi_t(q', \cdot) d\Pi_s(q, q') \quad (\text{i.e., } \Pi_{t+s} = \Pi_t \circ_{\kappa} \Pi_s)$$

— the Chapman–Kolmogorov equation, derived from the structure rather than assumed. A family $\{\Pi_t\}$ satisfying it is a *Markov semigroup kernel*. The associated *Markov operators* are

$$(K_t g)(q) := \int g d\Pi_t(q, \cdot), \quad g : \gamma \rightarrow \mathbb{R}.$$

Each K_t is positive, unital, and contractive on bounded measurable functions, with $K_0 = \text{id}$; positivity, unitality, and contractiveness are immediate from the Markov property.

Proposition 3.1 (Semigroup property). $K_{t+s}g = K_t(K_s g)$ for all bounded measurable g .

Proof. $(K_{t+s}g)(q) = \int g d(\Pi_t \circ_{\kappa} \Pi_s)(q, \cdot) = \int (\int g d\Pi_t(q', \cdot)) d\Pi_s(q, q') = \int (K_t g)(q') d\Pi_s(q, q') = (K_t(K_s g))(q)$, where the exchange of integrals is justified by Bochner–Fubini. $\square$

The pushforward of a measure μ on γ through the kernel is $P_t^* \mu := \mu \star \Pi_t = \int \Pi_t(q, \cdot) d\mu(q)$.

3.2 Koopman–Perron duality

Theorem 3.2 (Koopman–Perron duality). For every bounded measurable g and finite measure μ :

$$\int (K_t g) d\mu = \int g d(P_t^* \mu).$$

Proof. $\int g d(P_t^* \mu) = \int g d(\mu \star \Pi_t) = \int_{\gamma} \int g d\Pi_t(q, \cdot) d\mu(q) = \int (K_t g) d\mu$, by Bochner–Fubini. $\square$

When $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$ for a measurable map $\varphi_t : \gamma \rightarrow \gamma$, the stochastic picture collapses.

3.3 Specialisation to measure-preserving systems

Proposition 3.3 (Deterministic limit). *If $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$, then $K_t g = g \circ \varphi_t$ and $\varphi_{t+s} = \varphi_t \circ \varphi_s$.*

Proof. $(K_t g)(q) = \int g d\delta_{\varphi_t(q)} = g(\varphi_t(q))$. The flow law follows from $\delta_{\varphi_{t+s}(q)} = (\Pi_t \circ_\kappa \Pi_s)(q, \cdot) = \delta_{\varphi_t(\varphi_s(q))}$ and injectivity of the Dirac embedding. $\square$

In the deterministic case, $K_t g = g \circ \varphi_t = U_T^t g$: the Markov operator specialises to the classical Koopman operator. The Koopman–Perron duality $\int K_t g d\mu = \int g d(P_t^* \mu)$ becomes the change-of-variables formula $\int g \circ T^t d\mu = \int g d(T_*^t \mu)$.

Example 3.4 (Finite Markov chain). Let $X = \{1, \dots, k\}$, $\mathcal{B} = 2^X$, and let $P = (p_{ij})$ be the transition matrix of a finite irreducible Markov chain with unique stationary distribution μ ($\mu P = \mu$). Take full-state observation $Q_t = X_t$. The conditional regularity kernel of X_1 given X_0 is $\kappa(i, \cdot) = P(i, \cdot)$, the i -th row of P : this is the unique Markov kernel satisfying $\mathbb{E}_\mu[f(X_1) \mid X_0 = i] = \sum_j p_{ij} f(j)$ for all bounded f . The t -step kernel $\Pi_t(i, \cdot) = P^t(i, \cdot)$ is the corresponding semigroup kernel, and temporal coherence gives $\Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s$, i.e. $P^{t+s} = P^t P^s$. The semigroup acts as

$$K_t f(i) = \sum_j p_{ij}^t f(j).$$

Koopman–Perron duality reduces to the stationary-measure identity:

$$\int K_t g d\mu = \mu^\top P^t g = (\mu P^t)^\top g = \mu^\top g = \int g d\mu,$$

the fixed-point property $\mu P = \mu$ of the stationary distribution. In this finite setting every charge is automatically σ -additive, so the admissibility condition of [11] is trivially satisfied. Reconstruction holds at lag 0: $\mathcal{O}_h = \sigma(h) = \mathcal{B}(X)$ whenever h separates points of X .

The full structure produced by the construction is the triple $(\mathcal{P}(\beta), \nu_{Q_*}, \{K_t\}_{t \in \mathbb{N}})$: the canonical factor space, stationary distribution $\nu_{Q_*} = P \circ Q_*^{-1}$, and Markov semigroup. This is the *observable dynamical system* derived from (Q, F, P) : the conditional regularity of F relative to Q , promoted to a dynamical object by temporal coherence.

4 From regularity to reconstruction

Setting $Q_t = h \circ T^t$ in a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $h \in L^\infty$ specialises the construction. Under this choice, $Q_*(\omega) = (h(T^n \omega))_{n \in \mathbb{Z}}$ is the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$, and the observable dynamical system is the triple $(\Phi_h(X), \mu \circ \Phi_h^{-1}, \{K_t\})$.

The canonical factor space $\Phi_h(X)$ is a subset of $\mathbb{R}^\mathbb{Z}$ encoding the dynamical information that h can see. Whether it is a faithful image of X depends on how much the delay orbit $\{h \circ T^n : n \in \mathbb{Z}\}$ collectively distinguishes points of X . Injectivity of Φ_h modulo μ -null sets is equivalent to the observable σ -algebra

$$\mathcal{O}_h = \sigma(\{h \circ T^n : n \in \mathbb{Z}\})$$

equalling $\mathcal{B}(X)$ modulo μ .

Definition 4.1 (Faithful factor). A measurable map $\psi : (X, \mathcal{B}, \mu) \rightarrow (Y, \mathcal{Y})$ is *faithful modulo μ* if

$$\sigma(\psi) = \mathcal{B} \quad \text{modulo } \mu,$$

equivalently, if ψ separates μ -a.e. pair of points of X . The reconstruction question is whether Φ_h is faithful modulo μ .

The Koopman–Perron duality (Theorem 3.2) connects this to a function-space question: the observable σ -algebra $\mathcal{O}_h$ determines the Markov semigroup through the algebra it generates, and the reconstruction question is precisely whether that algebra is large enough to recover all of $\mathcal{B}$.

4.1 Separation defect and the role of CE

For a finite joined query algebra $\mathcal{G} = \mathcal{E}_{n_1} \vee \cdots \vee \mathcal{E}_{n_k}$ from a directed system $(\mathcal{E}_n)_n$, the *separation defect* is

$$\delta(\mathcal{G}) := \sup_{A \in \sigma(\mathcal{C})} \inf_{E \in \mathcal{G}} \mu(A \triangle E),$$

where $\sigma(\mathcal{C})$ is the observable σ -algebra of the full system. It vanishes iff $\sigma(\mathcal{G}) = \sigma(\mathcal{C}) \bmod \mu$. A sequence $(\mathcal{G}_k)$ is *honest* if it is increasing and $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C}) \bmod \mu$; under an honest scheme $\delta(\mathcal{G}_k)$ is non-increasing.

Theorem 4.2 (CE drives separation defect to zero). *Let $(\mathcal{E}_n)_n$ be a query system with realization measure μ satisfying collective exhaustion [11]. Under any honest refinement scheme $(\mathcal{G}_k)$,*

$$\delta(\mathcal{G}_k) \longrightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Proof. Fix $A \in \sigma(\mathcal{C})$. Since $(\mathcal{G}_k)$ is honest, $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C}) \bmod \mu$, so A is $\sigma(\bigcup_k \mathcal{G}_k)$ -measurable. By the $L^1(\mu)$ martingale convergence theorem, $\mathbf{1}_{E_k} \rightarrow \mathbf{1}_A$ in $L^1(\mu)$ where $E_k = \mathbb{E}_\mu[\mathbf{1}_A \mid \mathcal{G}_k]$ rounded to the nearest atom; in particular $\mu(A \triangle E_k) \rightarrow 0$. CE [11] ensures no mass persists on the vanishing sets $A \triangle E_k$, confirming $\inf_{E \in \mathcal{G}_k} \mu(A \triangle E) \rightarrow 0$ uniformly over A . Hence $\delta(\mathcal{G}_k) \rightarrow 0$. $\square$

Sections 5–6 carry out the reconstruction in the delay-map setting.

Remark 4.3 (The temporal specialisation). With $Q_t = h \circ T^t$, the algebra $\mathcal{O}_h^{(L)}$ is an honest refinement scheme once reconstruction holds (§§5–6). The separation defect specialises to $\delta(L) := \delta(\mathcal{O}_h^{(L)})$, the central quantity of [10]: exponential decay of $\delta(L)$ under mixing yields minimax-optimal certification rates. In the general query setting, exhaustion and mixing are separate conditions; the delay-map setting collapses the first by construction.

5 Density bridge and reconstruction

5.1 Density bridge lemma

Throughout this section and the next, $(X, \mathcal{B}, \mu)$ is a standard probability space, $T : X \rightarrow X$ is a measurable, μ -preserving invertible bimeasurable bijection with $T_*\mu = \mu$, and $h \in L^\infty(X, \mu)$.

Definition 5.1 (Observable algebra). The *observable algebra* generated by h is

$$\mathcal{O}_h := \sigma(\{h \circ T^n : n \in \mathbb{Z}\}) \subseteq \mathcal{B}(X).$$

The *delay algebra* $\mathcal{A}_h$ is the smallest subalgebra of $L^\infty(X, \mu)$ containing 1 and every $h \circ T^n$; concretely, finite sums $\sum_\alpha c_\alpha \prod_k (h \circ T^{n_k})^{e_k}$.

Definition 5.2 (Delay map). The *delay map* is

$$\Phi_h : X \rightarrow \mathbb{R}^{\mathbb{Z}}, \quad \Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}.$$

Since $\pi_n \circ \Phi_h = h \circ T^n$ for each coordinate projection π_n , the map Φ_h is measurable and $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. Thus $\mathcal{O}_h = \mathcal{B} \bmod \mu$ is precisely the condition that Φ_h is a measure-theoretic embedding.

The central lemma connects the Boolean algebra level to the L^2 level. It holds for any subalgebra of L^∞ .

Lemma 5.3 (Density bridge). *Let $(X, \mathcal{B}, \mu)$ be a probability space and let $\mathcal{A} \subseteq L^\infty(X, \mu)$ be a subalgebra containing the constant function 1. Then:*

$$\overline{\mathcal{A}}^{L^2} = L^2(X, \sigma(\mathcal{A}), \mu).$$

Consequently, $\overline{\mathcal{A}}^{L^2} = L^2(X, \mu)$ if and only if $\sigma(\mathcal{A}) = \mathcal{B}$ modulo μ .

Proof. Write $\mathcal{M} := \sigma(\mathcal{A})$ and $V := \overline{\mathcal{A}}^{L^2}$.

Step 1: $V \subseteq L^2(X, \mathcal{M}, \mu)$. Every $f \in \mathcal{A}$ is $\mathcal{A}$ -measurable, hence $\mathcal{M}$ -measurable. So $\mathcal{A} \subseteq L^2(X, \mathcal{M}, \mu)$. Since $L^2(X, \mathcal{M}, \mu)$ is a closed subspace of $L^2(X, \mathcal{B}, \mu)$, taking the closure gives $V \subseteq L^2(X, \mathcal{M}, \mu)$.

Step 2: $L^2(X, \mathcal{M}, \mu) \subseteq V$. It suffices to show that every indicator $\mathbf{1}_E$ with $E \in \mathcal{M}$ lies in V . By the functional monotone class theorem [4], the class

$$\mathcal{H} := \{f \in L^\infty(X, \mathcal{M}, \mu) : f \in V\}$$

is a vector space containing $\mathcal{A}$ and closed under bounded monotone pointwise limits (since if $f_n \in V$ are uniformly bounded and $f_n \rightarrow f$ pointwise a.e. with $f \in L^\infty$, then $f_n \rightarrow f$ in L^2 by dominated convergence, so $f \in V$). Since $\mathcal{A}$ is an algebra containing 1 and $\sigma(\mathcal{A}) = \mathcal{M}$, the monotone class theorem gives $\mathcal{H} = L^\infty(X, \mathcal{M}, \mu)$. In particular every $\mathcal{M}$ -indicator $\mathbf{1}_E \in V$.

Since the $\mathcal{M}$ -simple functions are dense in $L^2(X, \mathcal{M}, \mu)$ and V is closed, $L^2(X, \mathcal{M}, \mu) \subseteq V$.

Conclusion. Steps 1 and 2 give $V = L^2(X, \mathcal{M}, \mu)$. The final equivalence is immediate: $V = L^2(X, \mathcal{B}, \mu)$ iff $L^2(X, \mathcal{M}, \mu) = L^2(X, \mathcal{B}, \mu)$ iff $\mathcal{M} = \mathcal{B}$ mod μ . $\square$

Remark 5.4 (Role of the algebra structure). The algebra structure of $\mathcal{A}$ is essential. The corresponding statement for the closed linear span $\mathcal{H}_{\mathcal{A}} = \overline{\text{span}}(\mathcal{A})$ is false in general: $\mathcal{H}_{\mathcal{A}} = L^2$ does not imply $\sigma(\mathcal{A}) = \mathcal{B}$.

For example, take $X = [0, 1]$, $\mu = \text{Lebesgue}$, $T(x) = 2x \bmod 1$ (the doubling map), and $h(x) = x$. The orbit $\{h \circ T^n\}(x) = \{2^n x \bmod 1\}$ generates $\mathcal{B}([0, 1])$ as a σ -algebra (the dyadic structure separates points). But $\text{span}\{2^n x \bmod 1 : n \geq 0\}$ is a proper subspace of $L^2[0, 1]$: it does not contain x^2 . So the linear span can be a proper dense subspace even when $\sigma(\mathcal{A}) = \mathcal{B}$.

In the other direction, h cyclic for U_T (linear span dense) always implies $\sigma(\mathcal{A}_h) = \mathcal{B}$ mod μ — see Corollary 6.2.

5.2 Reconstruction theorem

Theorem 5.5 (Reconstruction theorem). *Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system and $h \in L^\infty(X, \mu)$. The following are equivalent:*

- (i) $\mathcal{O}_h = \mathcal{B}$ modulo μ ;
- (ii) $\mathcal{A}_h$ is dense in $L^2(X, \mu)$;
- (iii) $\Phi_h : X \rightarrow \mathbb{R}^\mathbb{Z}$ is a measure-theoretic embedding: $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z})) = \mathcal{B}$ modulo μ .

Proof. (i) $\Leftrightarrow$ (ii). Apply the density bridge (Lemma 5.3) with $\mathcal{A} = \mathcal{A}_h$ and $\sigma(\mathcal{A}_h) = \mathcal{O}_h$: $\overline{\mathcal{A}_h}^{L^2} = L^2(X, \mu)$ iff $\mathcal{O}_h = \mathcal{B}$ mod μ .

(i) $\Leftrightarrow$ (iii). By Definition 5.2, $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z}))$. So $\mathcal{O}_h = \mathcal{B}$ mod μ iff $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z})) = \mathcal{B}$ mod μ , which is exactly condition (iii). $\square$

Condition (iii) means Φ_h induces an isomorphism of measure algebras $(X/\mu, \mathcal{B}/\mu) \cong (\mathbb{R}^\mathbb{Z}/(\Phi_h)_*\mu, \mathcal{B}(\mathbb{R}^\mathbb{Z})/(\Phi_h)_*\mu)$. Since $(\Phi_h(Tx))_n = h(T^{n+1}x) = (\sigma(\Phi_h(x)))_n$, the measure $(\Phi_h)_*\mu$ is shift-invariant, and (X, T, μ) is isomorphic to $(\mathbb{R}^\mathbb{Z}, \sigma, (\Phi_h)_*\mu)$ restricted to $\Phi_h(X)$.

6 Cyclic vectors and Stone space identification

6.1 Cyclic vectors

Definition 6.1 (Cyclic vector). The function $h \in L^2(X, \mu)$ is a *cyclic vector* for U_T if

$$\mathcal{H}_h := \overline{\text{span}}\{U_T^n h : n \in \mathbb{Z}\} = L^2(X, \mu).$$

Corollary 6.2 (Cyclic implies reconstruction). *If $h \in L^\infty(X, \mu)$ is a cyclic vector for U_T , then $\mathcal{O}_h = \mathcal{B}$ modulo μ .*

Proof. Since $\mathcal{A}_h$ contains $\text{span}\{h \circ T^n\} = \text{span}\{U_T^n h\}$ as a subset, $\overline{\mathcal{A}_h}^{L^2} \supseteq \mathcal{H}_h = L^2$. By the density bridge, $\mathcal{O}_h = \mathcal{B} \bmod \mu$. $\square$

Remark 6.3 (The implication is strict). The converse of Corollary 6.2 fails in general: $\mathcal{O}_h = \mathcal{B}$ does not imply h is cyclic. The doubling map example of Remark 5.4 witnesses this.

The cyclic vector condition asks for density of the *linear span* of the orbit; reconstruction asks only for density of the *algebra*. For a bounded h , the linear span is contained in the algebra, so cyclicity is the stronger condition.

The two conditions coincide precisely when U_T has *simple L^2 -spectrum*: $L^2(X, \mu)$ is a cyclic U_T -module, meaning there exists some cyclic vector. In this case, every generating observation h (satisfying $\mathcal{O}_h = \mathcal{B}$) is automatically cyclic. Simple spectrum is a generic property of ergodic transformations in the Baire category sense [5] but is not universal.

6.2 Stone space identification

The Stone space $\text{St}(\mathcal{O}_h)$ of the observable algebra is the compact Hausdorff space whose points are ultrafilters on $\mathcal{O}_h$, with the natural Stone embedding $\iota : X \rightarrow \text{St}(\mathcal{O}_h)$ sending x to the principal ultrafilter $\{E \in \mathcal{O}_h : x \in E\}$.

Theorem 6.4 (Stone space identification). *Suppose $\mathcal{O}_h = \mathcal{B}$ modulo μ . Then ι is a measure-theoretic isomorphism: ι identifies the measure algebra $(X/\mu, \mathcal{B}/\mu)$ with $(\text{St}(\mathcal{O}_h)/\iota_*\mu, \mathcal{B}(\text{St}(\mathcal{O}_h))/\iota_*\mu)$.*

Proof. *Injectivity mod μ .* Suppose $\iota(x) = \iota(x')$. For any $E \in \mathcal{B}$ with $x \in E \not\equiv x'$, find $F \in \mathcal{O}_h$ with $\mu(E \triangle F) = 0$. Since $x \in F$ and $\iota(x) = \iota(x')$, also $x' \in F$, so $x' \in (E \triangle F) \cup F^c$, placing x' in a null set. Hence ι is injective μ -a.e.

σ -algebra identification. The pullback ι^{-1} maps the Baire σ -algebra of $\text{St}(\mathcal{O}_h)$ (generated by the clopen sets $[E] = \{u : E \in u\}$) to $\mathcal{O}_h = \mathcal{B} \bmod \mu$.

Measure recovery. Since μ is σ -additive on $\mathcal{O}_h = \mathcal{B}$, the collective exhaustion condition of [11] forces $\iota_*\mu$ to concentrate on the principal ultrafilters $\iota(X)$, so $\iota_*\mu$ is supported on the image and the measure algebras are identified. $\square$

The Stone space [11] — built as a compact ambient space for deriving σ -additivity — is, under the reconstruction condition, the state space X itself. The construction that began by seeking a compact completion of the sample space ends by recognising it as the object the observer was trying to recover.

Example 6.5 (Cantor set: reconstruction without smoothness). Let $\mathcal{C} \subset [0, 1]$ be the middle-thirds Cantor set, and let $\mu_{\mathcal{C}}$ be the Cantor measure — the pushforward of the fair-coin product measure $\nu = (\frac{1}{2}\delta_0 + \frac{1}{2}\delta_2)^{\mathbb{N}}$ under the ternary coding map $\pi(d_1, d_2, \dots) = \sum_{k \geq 1} d_k/3^k$. Let $S : \mathcal{C} \rightarrow \mathcal{C}$ be the shift map

$$S(x) = \begin{cases} 3x & x \in \mathcal{C} \cap [0, \frac{1}{3}], \\ 3x - 2 & x \in \mathcal{C} \cap [\frac{2}{3}, 1], \end{cases}$$

and let $h(x) = x$. The system $(\mathcal{C}, \mu_{\mathcal{C}}, S)$ is measurably conjugate to the Bernoulli shift $(\{0, 2\}^{\mathbb{N}}, \nu, \sigma)$ via π ; in particular S is $\mu_{\mathcal{C}}$ -invariant and ergodic.

For $x = \pi(d_1, d_2, \dots)$ we have $S^n x = \pi(d_{n+1}, d_{n+2}, \dots)$. Since every point of $\mathcal{C}$ lies in $[0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$,

$$d_{n+1} = 0 \iff S^n x \in [0, \frac{1}{3}], \quad d_{n+1} = 2 \iff S^n x \in [\frac{2}{3}, 1],$$

so $h \circ S^n$ recovers the $(n+1)$ -st ternary digit of x . The delay vector $\Phi_h^{(L)}(x) = (x, Sx, \dots, S^L x)$ reads off digits $d_1, \dots, d_{L+1}$; its fibres are the generation- $(L+1)$ cylinder sets

$$F_z = \{x \in \mathcal{C} : d_k(x) = z_k, 1 \leq k \leq L+1\}, \quad z \in \{0, 2\}^{L+1},$$

each carrying $\mu_{\mathcal{C}}(F_z) = 2^{-(L+1)}$, so $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)}) = \sigma\{F_z : z \in \{0, 2\}^{L+1}\}$. Since $\mu_{\mathcal{C}}$ -a.e. pair of distinct points differs at some finite digit, these algebras separate points as $L \rightarrow \infty$:

$$\mathcal{O}_h = \sigma\left(\bigcup_{L \geq 0} \mathcal{O}_h^{(L)}\right) = \mathcal{B}(\mathcal{C}) \mod \mu_{\mathcal{C}}.$$

By Theorem 5.5, Φ_h is a measure-theoretic embedding: the finite-lag observable algebras refine to the full Borel structure as $L \rightarrow \infty$.

The observable algebra also encodes the geometry of $\mathcal{C}$ directly. At each lag L there are 2^{L+1} fibres of Euclidean diameter $(1/3)^{L+1}$, so the box-counting ratio $\log 2^{L+1} / \log 3^{L+1} = \log 2 / \log 3$ is an algebraic invariant of the algebra itself, constant across all lags. The Hausdorff dimension is not a separate geometric calculation: it is read off from the refinement rate of the observable algebra.

$\mathcal{C}$ is not a smooth manifold: it is compact and zero-dimensional, with Hausdorff dimension $\log 2 / \log 3 \approx 0.63$. The differential-geometric hypotheses of Takens's theorem — C^2 ambient dynamics, integer manifold dimension d , embedding bound $2d+1$ — are unavailable before one even asks about regularity of S ; nor do the finite-lag extensions of Sauer–Yorke–Casdagli [13], which still require a finite-dimensional ambient manifold or attractor, apply. Theorem 5.5 applies because it requires only that the observable algebra be measurably generated; smoothness and finite dimensionality play no role.

The classical Takens embedding theorem [14] reaches a related conclusion by a different route: for a C^2 diffeomorphism on a compact d -manifold and a generic smooth h , the finite-delay map $\Phi_h^{(k)}(x) = (h(x), h(Tx), \dots, h(T^{k-1}x))$ is a diffeomorphic embedding into $\mathbb{R}^{2d+1}$ for $k \geq 2d+1$. Sauer–Yorke–Casdagli [13] extend this to prevalence-generic observables and Lipschitz maps, while retaining the finite-dimensional manifold hypothesis and the finite-lag, finite-dimension embedding framework. The reconstruction theorem here differs in kind: it requires only measurability, uses all delays simultaneously, and replaces the dimension count with the algebraic density condition; in exchange, the embedding is measure-theoretic rather than topological. On a compact manifold with generic smooth h , the orbit $\{h \circ T^n\}$ separates points, so Stone–Weierstrass gives $\mathcal{A}_h$ uniformly dense in $C(M)$ and hence L^2 -dense: the density condition holds and the results are compatible. For non-smooth or zero-dimensional systems — the Cantor example being representative — the closer historical ancestor is the symbolic dynamics and generating-partition tradition [1]: representing orbits through successively finer finite partitions is exactly what the observable algebra $\mathcal{O}_h^{(L)}$ does as L grows, with faithfulness of Φ_h replacing the generating-partition condition.

Remark 6.6 (Observable separation and Lyapunov exponents). When T is C^r and reconstruction holds, the delay map Φ_h carries a natural instability rate: for μ -a.e. x and nearby y ,

$$\lambda_h(x, y) := \limsup_{n \rightarrow \infty} \frac{1}{n} \log d(\Phi_h(T^n x), \Phi_h(T^n y)).$$

This *observable separation exponent* measures how fast the reconstructed dynamics distinguishes initial conditions through the observable, rather than how fast tangent vectors grow. When h is generic in the sense of Takens [14], the finite-lag embedding $\Phi_h^{(L)}$ is bi-Lipschitz on the attractor, sandwiching λ_h between the top Lyapunov exponent λ_1 from above and below; one expects $\lambda_h = \lambda_1$ μ -a.e., but making this precise requires comparing the induced metric on $\Phi_h^{(L)}(X)$ with the ambient Riemannian geometry as $L \rightarrow \infty$, which is beyond the scope of the present paper. The quantitative geometry of fibre separation in this setting is taken up in [10].

7 Discussion

The measure derived in [11] already contains, without further assumption, the structures developed here. The Rokhlin disintegration reads off a conditional regularity kernel for any pair of observables; temporal indexing of that kernel forces Chapman–Kolmogorov and hence the Markov semigroup; and in a measure-preserving system the minimal sufficient factor becomes the delay map. Nothing is added to the measure — only consequences are drawn.

The mathematical content of §§2–6 is standard once a σ -additive measure is in hand on the observable algebra; what the programme changes is the explanatory position of each object. The individual components are classical. Rokhlin disintegration [12] gives regular conditional distributions on standard Borel spaces; Halmos and Savage [6] laid the measure-theoretic foundation for sufficient statistics via the Radon–Nikodym theorem; Blackwell [2] placed sufficiency in comparative terms; and Doob [3] established conditional expectation, filtrations, and Markov semigroups as the language of modern probability. Predictive-state representations [9] and observable operator models [7] likewise distrust hidden-state parametrisations and encode state through observable predictions. Here the kernel, the factor, the semigroup, and the delay map are not assumed, chosen, or built from a prior dynamical law — they are consequences of a single measure on an observable algebra, disclosed in sequence.

The reconstruction theorem closes a circle opened in [11]. The Stone space $\text{St}(\mathcal{C})$ built there to establish σ -additivity is an abstract completion of the observable algebra, containing the realised state space as the principal ultrafilters but not yet identified with it. When Φ_h is faithful modulo μ (Definition 4.1), the three equivalent conditions of Theorem 5.5 jointly force $\sigma(\Phi_h) = \mathcal{B}$ modulo μ , and the Stone space collapses onto X itself: the abstract completion is the concrete state space. The question left open in [11] — whether prediction, dynamics, and state recovery follow from observation — is answered affirmatively, under faithfulness.

What faithfulness means in practice, and whether it can be certified from finite data, is taken up in [10].

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